

Centered MLR Window Sensitivity for VJ146A2 Eskom Validation

Whole-hour windows around the selected 01:00-02:00 overpass specification

Reliable Electrification Planning SA

2026-06-04

Purpose

This note checks whether the selected VJ146A2/Eskom validation is sensitive to the Eskom MLR exposure window. It is intentionally separate from the main validation report.

The preferred specification uses Eskom MLR in the 01:00-02:00 SAST overpass window. This sensitivity keeps the same standard validation sample and compares it with two whole-hour windows around that nighttime period.

This version uses the corrected MLR share denominator:

$$\frac{\sum \text{Manual Load Reduction}}{\sum \text{RSA Contracted Demand}}$$

The Eskom glossary defines MLR as reduced demand due to load shedding and/or curtailment, while RSA Contracted Demand is contracted demand rather than served demand excluding MLR. The earlier selected-specification column used the legacy denominator $MLR/(MLR + RSA \text{ Contracted Demand})$, so this report keeps a provenance check for the legacy value but reports corrected regressions.

Table 1: Whole-hour MLR window definitions used in this sensitivity.

exposure	hours_relative_to_local_overpass_date	interpretation
01:00-02:00 preferred	1	Selected overpass-hour exposure used in the recommended report.
23:00-03:00 centered 4h	-1,0,1,2	Whole-hour 4-hour sensitivity around the overpass hour, using 23:00 previous day through 02:00 on local_overpass_date.
13:00-13:00 centered 24h	-11,-10,-9,-8,-7,-6,-5,-4,-3,-2,-1,0,1,2,3,4,5,6,7,8,9,10,11,12	Whole-hour 24-hour sensitivity around the nighttime period, using 13:00 previous day through 12:00 on local_overpass_date.

The sample is unchanged: 267 standard validation dates after the demand-gate screen and annual-IQR excess-lit removal. The legacy recomputation for 01:00-02:00 matches the stored selected-specification column with maximum absolute difference 5e-16. The corrected denominator differs from the stored legacy selected column by as much as 2.63 percentage points.

Window Results

Table 2: OLS validation results by centered whole-hour MLR window. The MLR regressor is $\text{sum(MLR)} / \text{sum(RSA Contracted Demand)}$. Standard errors use Newey-West HAC with 7-day lag.

Window	Outcome	Mean MLR/contracted	Effect per +10pp MLR/contracted	HAC SE	p- value	Pearson r	R2
01:00-02:00 preferred	Popw strict dark	5.9%	2.42 pp	0.14 pp	7.79e- 45	0.762	0.581
23:00-03:00 centered 4h	Popw strict dark	6.2%	2.60 pp	0.15 pp	6.95e- 44	0.761	0.580
13:00-13:00 centered 24h	Popw strict dark	7.3%	2.37 pp	0.15 pp	7.77e- 39	0.735	0.540
01:00-02:00 preferred	Popw mostly dark	5.9%	4.05 pp	0.26 pp	4.75e- 40	0.745	0.555
23:00-03:00 centered 4h	Popw mostly dark	6.2%	4.37 pp	0.28 pp	1.40e- 39	0.746	0.557
13:00-13:00 centered 24h	Popw mostly dark	7.3%	4.02 pp	0.29 pp	1.01e- 33	0.725	0.526
01:00-02:00 preferred	Demandw strict dark	5.9%	3.43 pp	0.19 pp	6.87e- 48	0.786	0.618
23:00-03:00 centered 4h	Demandw strict dark	6.2%	3.70 pp	0.21 pp	8.17e- 46	0.785	0.616
13:00-13:00 centered 24h	Demandw strict dark	7.3%	3.37 pp	0.21 pp	7.34e- 42	0.757	0.573
01:00-02:00 preferred	Demandw mostly dark	5.9%	5.27 pp	0.30 pp	2.30e- 46	0.779	0.608
23:00-03:00 centered 4h	Demandw mostly dark	6.2%	5.70 pp	0.33 pp	3.68e- 45	0.781	0.610
13:00-13:00 centered 24h	Demandw mostly dark	7.3%	5.22 pp	0.33 pp	1.70e- 39	0.757	0.573

Centered Eskom MLR windows preserve the VJ darkness signal

OLS slopes use Newey–West HAC standard errors with 7-day lag; same 267 standard validation days.

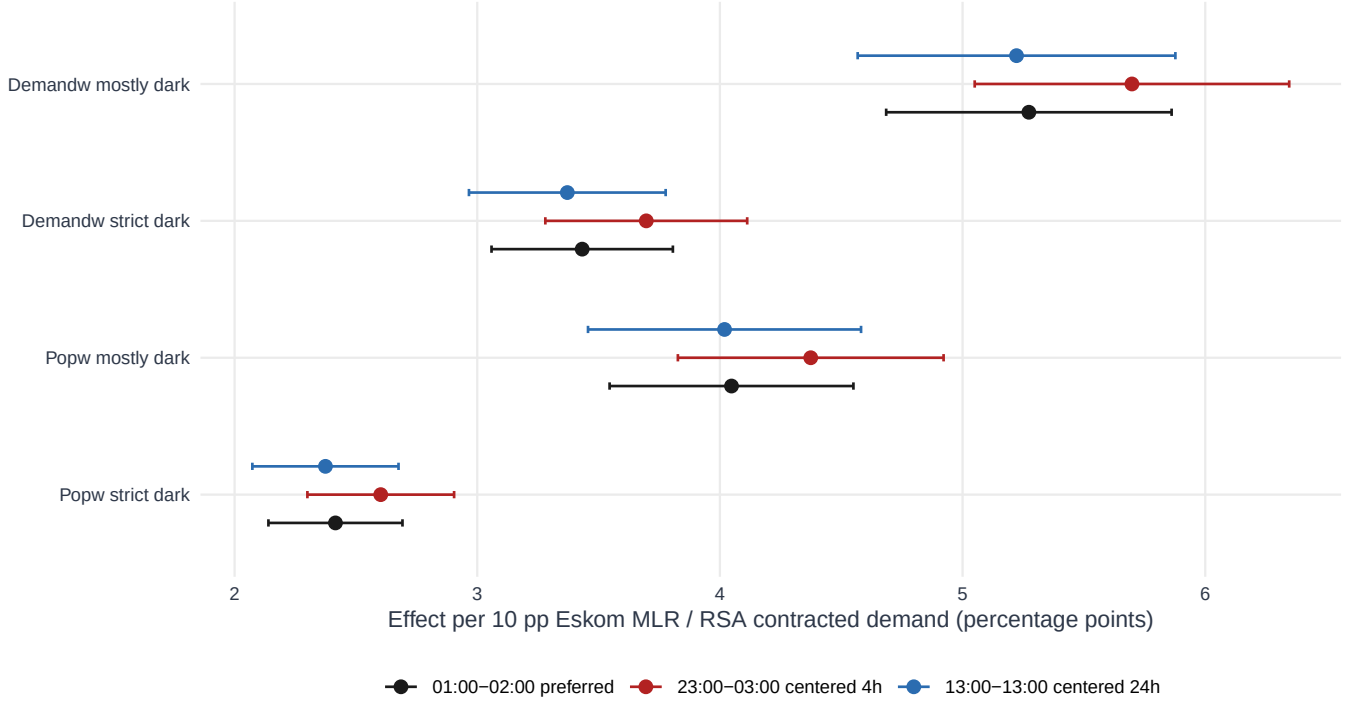


Figure 1: Window-sensitivity coefficients for the main darkness outcomes.

The centered 4-hour window is not weaker than the preferred hour. For population-weighted strict darkness, the preferred 01:00–02:00 model has $R^2 = 0.581$, while the centered 4-hour model has $R^2 = 0.580$. The centered 24-hour model remains strong, with $R^2 = 0.540$, but it is weaker than the nighttime windows.

Mean MLR MW Specification

This table uses the simple average MLR MW inside each fixed window:

$$\frac{1}{H} \sum_h \text{Manual Load Reduction}_h$$

For a fixed window, the mean-MW and sum-MWh regressors contain the same information up to a constant scale. Reporting the mean is useful because it is directly comparable to hourly PyPSA power quantities.

Table 3: OLS validation results using mean MLR MW inside each centered whole-hour window. Standard errors use Newey–West HAC with 7-day lag.

Window	Outcome	Mean MLR MW	Effect per +1,000 MW mean MLR	HAC SE	p-value	Pearson r	R2
01:00–02:00 preferred	Popw strict dark	1282	1.09 pp	0.07 pp	1.23e-42	0.760	0.578
23:00–03:00 centered 4h	Popw strict dark	1378	1.15 pp	0.07 pp	3.37e-41	0.759	0.576
13:00–13:00 centered 24h	Popw strict dark	1904	0.89 pp	0.06 pp	5.80e-41	0.738	0.544

Window	Outcome	Mean MLR MW	Effect per +1,000 MW mean MLR	HAC SE	p-value	Pearson r	R2
01:00-02:00 preferred	Popw mostly dark	1282	1.83 pp	0.12 pp	2.92e- 39	0.745	0.555
23:00-03:00 centered 4h	Popw mostly dark	1378	1.94 pp	0.13 pp	2.30e- 38	0.746	0.557
13:00-13:00 centered 24h	Popw mostly dark	1904	1.51 pp	0.10 pp	2.97e- 36	0.730	0.533
01:00-02:00 preferred	Demandw strict dark	1282	1.55 pp	0.09 pp	7.46e- 45	0.782	0.612
23:00-03:00 centered 4h	Demandw strict dark	1378	1.63 pp	0.10 pp	1.76e- 42	0.781	0.609
13:00-13:00 centered 24h	Demandw strict dark	1904	1.26 pp	0.07 pp	1.18e- 43	0.757	0.574
01:00-02:00 preferred	Demandw mostly dark	1282	2.38 pp	0.14 pp	1.28e- 44	0.778	0.605
23:00-03:00 centered 4h	Demandw mostly dark	1378	2.52 pp	0.15 pp	5.99e- 43	0.779	0.606
13:00-13:00 centered 24h	Demandw mostly dark	1904	1.95 pp	0.12 pp	6.46e- 42	0.759	0.576

Timing Figures

Daily Eskom MLR and VJ strict darkness: 01:00–02:00 preferred

Sample: 267 standard validation days; both series standardized over the same days.
Gray ticks show excluded dates. $r = 0.762$, $R^2 = 0.581$.



Figure 2: Daily standardized timing comparison for the preferred 01:00-02:00 window.

Daily Eskom MLR and VJ strict darkness: 23:00–03:00 centered 4h

Sample: 267 standard validation days; both series standardized over the same days.

Gray ticks show excluded dates. $r = 0.761$, $R^2 = 0.580$.



Figure 3: Daily standardized timing comparison for the centered 23:00-03:00 four-hour window.

Daily Eskom MLR and VJ strict darkness: 13:00–13:00 centered 24h

Sample: 267 standard validation days; both series standardized over the same days.

Gray ticks show excluded dates. $r = 0.735$, $R^2 = 0.540$.



Figure 4: Daily standardized timing comparison for the centered 13:00-13:00 twenty-four-hour window.

Baseline-Relative Dose Response

MLR windows show a similar baseline-relative darkness gradient

Zero is the mean of no-MLR standard validation nights inside each window; positive-MLR nights are split into ten equal-count bins.

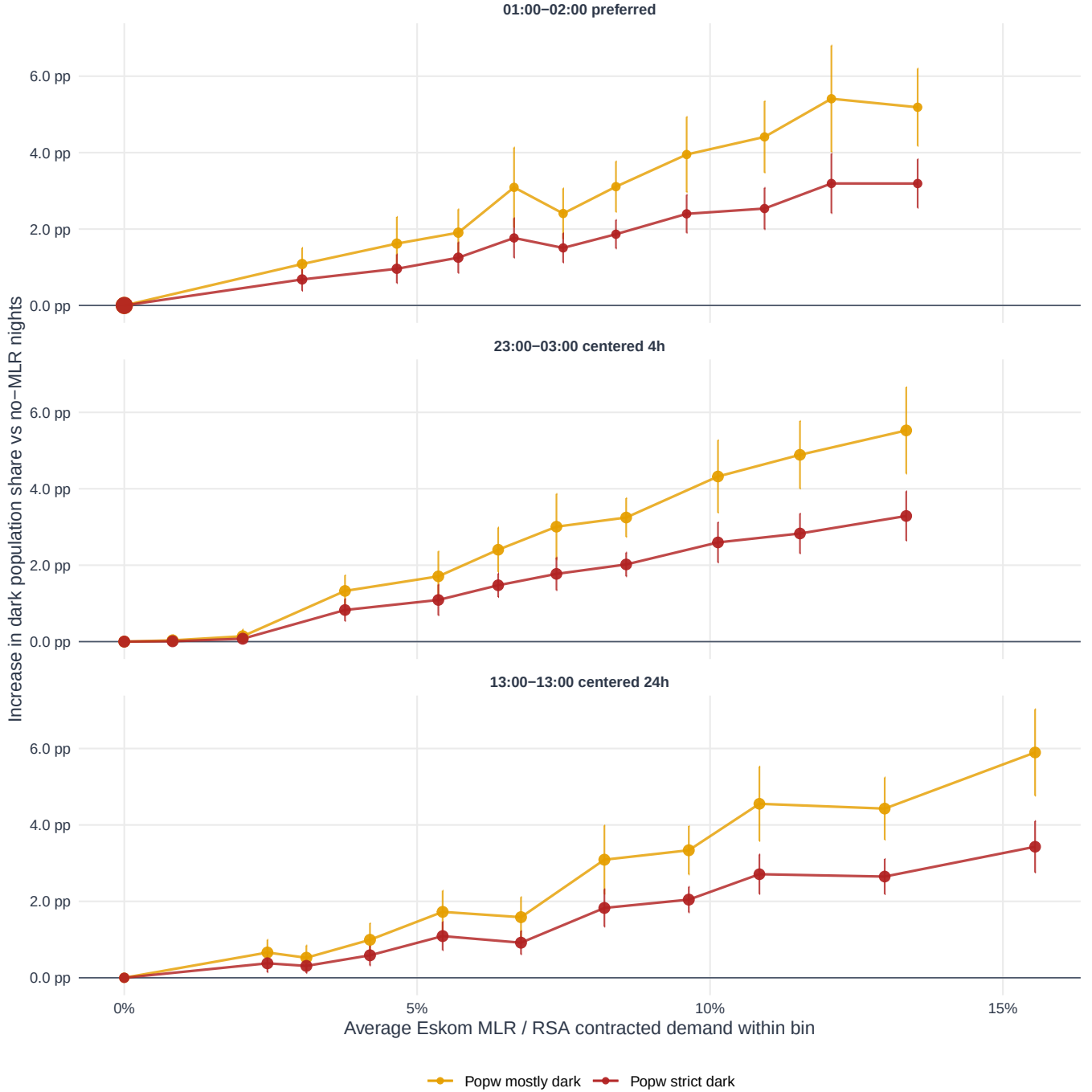


Figure 5: Baseline-relative darkness by centered MLR window. Zero is the mean of no-MLR standard validation nights inside each window.

The baseline-relative plots report magnitudes rather than standardized co-movement. Each panel computes its own no-MLR baseline for the corresponding exposure window, then splits positive-MLR nights into ten equal-count bins.

Table 4: Highest-MLR-bin baseline-relative endpoint by window.

Window	Outcome	Highest-bin mean MLR/contracted	Increase vs no-MLR	Baseline nights	Positive-bin nights
01:00-02:00 preferred	Popw mostly dark	13.5%	5.2 pp	74	19
01:00-02:00 preferred	Popw strict dark	13.5%	3.2 pp	74	19
23:00-03:00 centered 4h	Popw mostly dark	13.3%	5.5 pp	25	24
23:00-03:00 centered 4h	Popw strict dark	13.3%	3.3 pp	25	24
13:00-13:00 centered 24h	Popw mostly dark	15.5%	5.9 pp	20	24
13:00-13:00 centered 24h	Popw strict dark	15.5%	3.4 pp	20	24

Full-Day ENS-Like Metric

For the centered 24-hour window, the MLR sum can be read as an hourly energy-not-served proxy in MWh. The average centered 24-hour MLR burden in this selected sample is 45.70 GWh and the maximum is 125.56 GWh.

Table 5: Centered 24-hour MLR energy proxy models.

Outcome	Effect per +10 GWh MLR	HAC SE	p-value	Pearson r	R2
Popw strict dark	0.37 pp	0.02 pp	5.80e-41	0.738	0.544
Popw mostly dark	0.63 pp	0.04 pp	2.97e-36	0.730	0.533
Demandw strict dark	0.52 pp	0.03 pp	1.18e-43	0.757	0.574
Demandw mostly dark	0.81 pp	0.05 pp	6.46e-42	0.759	0.576

Interpretation

The sensitivity supports the selected validation rather than undermining it. The strongest results remain near the nighttime overpass period, but the centered 24-hour window also explains a large share of variation in the VJ darkness outcomes.

For the PyPSA-facing use case, the cautious interpretation is:

VJ-derived uptime is validated most directly against nighttime load reduction, but it also tracks same-day load-shedding burden over a centered 24-hour window. It can therefore be used as a spatial proxy for daily ENS intensity, while PyPSA or Eskom hourly profiles should provide the hourly allocation shape.

This note should be treated as a sensitivity candidate for the main validation report, not as a replacement for the preferred 01:00-02:00 specification.